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Plant Disease Classification with Deep Learning and LLM Support

CCAI323 - Machine Learning

Fall 2026

Team Members and Solution Assignments

Members	Student ID	Assigned Solution
Jude Alqhtani	2310397	using EfficientNet-B0 model for classification and applying LangChain
Waad Alhenaki	2310896	Using hybrid Supervised and Unsupervised models with Autoencoder and SVM for classification and applying LangChain
Lana Almalki	2311539	Using ConvNext–CBAM model for image classification tasks and applying LangChain
Aryam Almutairi	2310631	Using Mask R-CNN for image segmentation and classification tasks and applying LangChain.
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1. Executive Summary

To achieve a comprehensive and fair comparative evaluation, four distinct solutions were implemented alongside a State-of-the-Art (SOA) benchmark, each targeting a different dimension of the problem. Solution 1 (EfficientNet-B0), implemented by Jude Alqhtani, employs a lightweight Convolutional Neural Network with compound scaling, leveraging frozen ImageNet weights and a custom classification head to achieve high accuracy with minimal computational cost. Solution 2 Autoencoder + SVM, implemented by Waad Alhenaki, combines a Convolutional Autoencoder for unsupervised feature extraction with a Support Vector Machine classifier, forming a hybrid pipeline optimized for rapid inference and deployment in resource-constrained environments. Solution 3 (ConvNeXt-CBAM), implemented by Lana Almalki, applies a ConvNeXt-Tiny backbone enhanced with a Convolutional Block Attention Module (CBAM), directing the model's focus toward disease-relevant leaf regions through channel and spatial attention mechanisms. Solution 4 Mask R-CNN, implemented by Aryam Almutairi, extends disease detection beyond classification to include pixel-level segmentation and disease region localization using a ResNet-50 backbone with Feature Pyramid Networks. The SOA model PLA-ViT, led by Aseel Almatrafi, replicates the Vision Transformer architecture from Narayanan et al. (2025), leveraging self-attention mechanisms to capture global spatial dependencies across leaf image patches for state-of-the-art classification performance.
All five models were trained and evaluated on the same preprocessed dataset using a unified pipeline that included resizing to 224×224 pixels, ImageNet normalization, dynamic data augmentation random horizontal flip, rotation up to 15°, and color jitter, and an 80:20 train-validation split. This standardized setup ensures that performance differences across solutions reflect genuine architectural and methodological distinctions rather than preprocessing variation.

The key results of this project demonstrate that the SOA model PLA-ViT achieved the highest classification accuracy at 98.52%, confirming the superiority of self-attention mechanisms in capturing complex disease patterns. ConvNeXt-CBAM followed closely at 97.17%, offering a strong accuracy-efficiency balance with the added benefit of attention-guided predictions. EfficientNet-B0 reached 95.20% with the fastest training time and smallest computational footprint 4.01M parameters, 827.74M FLOPs, making it the most deployment-friendly solution. Mask R-CNN achieved 95.50% while uniquely providing pixel-level segmentation outputs alongside classification. The Autoencoder + SVM pipeline achieved 93.36% accuracy with the fastest inference speed at 3.23 ms per image, establishing it as the optimal choice for real-time agricultural monitoring applications. All four solutions integrated LangChain to provide AI-driven treatment recommendations upon disease identification, enhancing the practical usability of the system for end users such as farmers and agronomists.

The major challenges encountered across the project included the pronounced class imbalance in the dataset where the Grape_healthy class comprised only 423 samples compared to over 1,000 in each disease class which required careful augmentation and weighted evaluation strategies. Computational resource limitations posed challenges particularly for the ViT and Mask R-CNN architectures, addressed through transfer learning and pre-trained weight initialization. A data quality issue involving misaligned segmentation mask filenames was identified and resolved through a custom preprocessing script, ensuring pixel-level correspondence for the Mask R-CNN and Autoencoder solutions. Finally, adapting each architecture's classification head to the four-class problem and tuning hyperparameters within the 10-epoch training constraint required iterative experimentation across all solutions.

Overall, this project shows that no single model is best in all aspects. The optimal choice depends on the application needs, such as accuracy ViT, precision ConvNeXt-CBAM, efficiency EfficientNet-B0, localization Mask R-CNN, or real-time speed Autoencoder + SVM.

2. Introduction

2.1 Problem Description

Grape crops are highly susceptible to Black Rot, Esca (Black Measles), and Leaf Blight, which cause severe yield losses if undetected. Traditional manual inspection is costly and error-prone, especially when disease symptoms appear visually similar in early stages. This project addresses automated classification of grape leaf images into four classes Healthy, Black Rot, Esca, and Leaf Blight using the PlantVillage dataset, complicated further by class imbalance and the visual overlap between Black Rot and Esca.

2.2 Motivation and Significance

Early detection enables targeted treatment before widespread crop damage, reducing pesticide overuse and economic losses. Automating this process removes dependency on scarce agricultural expertise and enables scalable real-time monitoring via mobile and drone-based systems. This work also provides a valuable testbed for comparing fundamentally different ML paradigms under identical experimental conditions.

2.3 Target Audience and Impact

Primary beneficiaries are farmers, agronomists, and agricultural organizations seeking cost-effective early detection tools. This work additionally contributes to the research community by offering a rigorous multi-model comparison under a unified framework.

2.4 Project Objectives and Contributions

- Implement and evaluate four ML solutions covering supervised, hybrid, attention-based, and segmentation approaches.
- Replicate the state-of-the-art PLA-ViT model as a rigorous performance benchmark.
- Conduct comparative analysis with scientific ML justification for performance differences.
- Integrate LangChain-based LLM support to generate natural language treatment recommendations from model predictions.

3. Literature Review

3.1 Overview of Existing Approaches

Plant disease detection has evolved significantly over the past decade, transitioning from manual inspection and handcrafted image processing to highly automated deep learning frameworks. The landscape of existing solutions can be organized into four major thematic categories: classical machine learning approaches, CNN-based deep learning models, transformer and segmentation architectures, and unsupervised learning integrated with Large Language Models (LLMs).

Classical ML approaches combine CNN-extracted features with classifiers like Support Vector Machines SVMs and K-Nearest Neighbors KNN. Mohameth et al. (2020) demonstrated that combining ResNet50 and VGG16 features with SVM on the PlantVillage dataset achieves 97.82% accuracy, consistently outperforming KNN [1]. However, these hybrid pipelines depend on manual feature engineering and exhibit limited suitability for real-time edge deployment.

CNNs approaches dominate recent literature due to their ability to automatically learn hierarchical features. Ali et al. (2024) presented a fine-tuned EfficientNet-B0 model that achieved 99.78% accuracy on PlantVillage, outperforming architectures like Inception-v3 and ResNet50 [2]. Complementarily, Saddami et al. (2024) conducted a green AI study confirming that EfficientNet-B0 achieves 99.8% accuracy while maintaining the lowest computational footprint among lightweight CNNs [3].

Transformers & Segmentation architectures have gained attention for capturing global contextual relationships within images. Murugavalli and Gopi (2024) proposed PLA-ViT, utilizing self-attention mechanisms to outperform CNN benchmarks in detection accuracy, disease localization, and inference time [4]. Concurrently, detection and segmentation frameworks like Mask R-CNN extend classification by providing pixel-level outputs. Almazaydeh et al. (2022) applied Mask R-CNN for herbal leaf recognition, achieving 95.7% accuracy while generating precise bounding boxes and segmentation masks [5].

Unsupervised and self-supervised approaches have emerged to address the lack of labeled data. Benfenati et al. (2023) showed that autoencoder-based anomaly detection holds significant discriminative potential for powdery mildew on cucumber leaves [6]. Similarly, Chen et al. (2020) introduced SimCLR, a contrastive framework achieving performance competitive with fully supervised training using only 1% of labels [7].

Large Language Models (LLMs) have recently entered the agricultural domain. Samuel et al. (2025) developed AgroLLM, achieving 93% accuracy on agricultural Q&A tasks [8]. Another study (2025) found that fine-tuned GPT-4o reaches 98.12% accuracy on apple leaves, while zero-shot GPT-4o remains inferior to ResNet-50. This confirms that LLMs still require external vision models for competitive performance [9].

3.2 Relevant ML Techniques in Literature

Convolutional Neural Networks (CNNs): EfficientNet-B0 is a dominant choice due to its compound scaling strategy balancing depth, width, and resolution. Its success with ImageNet-pretrained weights across literature directly motivates its adoption as **Solution 1** in this project.

Unsupervised and Hybrid Pipelines: Autoencoders learn compact latent representations of healthy leaves, identifying diseased samples through reconstruction error [6]. SimCLR maximizes agreement between differently augmented views of the same image, requiring no labels during representation learning [7]. Combined with SVMs, which create robust class boundaries from deep features [1], these techniques form Solution 2, specifically targeting scenarios with limited labeled data.

ConvNeXt: This architecture modernizes traditional CNNs using transformer-inspired choices like larger depthwise kernels and LayerNorm. Wu et al. [10] showed that an improved ConvNeXt significantly outperforms standard CNNs on soybean leaf diseases. Solution 3 builds on this by applying ConvNeXt to multi-class plant disease classification.

Mask R-CNN: Extending Faster R-CNN with a pixel-level segmentation branch, this architecture simultaneously produces class predictions and instance masks. Almazaydeh et al. [5] demonstrated its viability for fine-grained leaf analysis. Solution 4 applies this framework to precise disease localization and severity estimation.

Vision Transformers (ViTs): ViTs use self-attention to model long-range dependencies across image patches, capturing the global structure of leaf images better than CNNs [4]. In this project, ViT serves as the SOA Benchmark to objectively evaluate our four solutions.

3.3 Gaps and Limitations

Despite the strong performance reported across the reviewed studies, several critical limitations collectively motivate our comparative framework:

- Lack of Actionable Agronomic Solutions (The Primary Gap): The vast majority of existing vision frameworks [1-7, 10] stop entirely at disease classification or localization. They successfully identify the disease but fail to provide actionable mitigation strategies. Our project directly bridges this gap: once our core models classify a specific grape leaf disease, an integrated LLM queries a knowledge base to transform a simple diagnostic label into a comprehensive, real-time treatment protocol.
- Single-Crop Generalization Gap: Most studies evaluate models on narrow datasets [2, 5, 10]. This limits confidence in model robustness. Our framework ensures rigorous evaluation across diverse grape leaf diseases.
- Under-Reported Computational Efficiency: Many studies prioritize accuracy over inference latency [2, 4]. Following green AI principles [3], our project explicitly benchmarks computational efficiency alongside accuracy.
- Underexplored Unsupervised Approaches: Despite the potential of SimCLR [7] and autoencoders [6], these remain underutilized in plant disease applications. Solution 2 directly addresses this gap.
- Absence of Standardized Benchmarks: Direct cross-architecture comparisons are rare. This project unifies CNNs, Hybrid pipelines, Mask R-CNN, and ViT under an identical experimental framework for objective evaluation.

3.4 Summary of Key Insights

The reviewed literature directly shaped our system design through several actionable insights:

- Diagnosis Must Pair with Treatment: The literature reveals a disconnect between detection and actionable solutions. This drives our use of an LLM as a post-classification engine to provide automated treatments.
- Transfer Learning & Augmentation: ImageNet pre-training and robust augmentation (e.g., masking, rotation) are non-negotiable for maximizing generalization on agricultural datasets.
- Architecture Selection: Convergent evidence establishes EfficientNet-B0 as the optimal accuracy-efficiency tradeoff Solution 1. The combination of SVM, Autoencoders, and SimCLR forms a robust pipeline for label-scarce scenarios Solution 2. ConvNeXt's modernized design provides measurable feature extraction gains Solution 3, while Mask R-CNN enables spatial localization beyond pure classification Solution 4. Finally, ViT's global attention mechanism makes it the ideal SOA baseline for comparative analysis.

Table 1: Summary of reviewed papers, assigned solutions, and key findings

Reference	Solution	Method / Approach	Key Result	Scalability	Limitations
Murugavalli & Gopi (2024)	SOA Benchmark: ViT	PLA-ViT: Vision Transformer with self-attention, data augmentation, bilateral filtering.	Outperforms CNN-based models in accuracy, disease localization, inference time & computational complexity	Moderate	High compute; requires large labeled data; struggles on small domain-shifted datasets
Ali et al. (2024)	Sol. 1: EfficientNet-B0	Fine-tuned EfficientNet-B0 with GMP, dropout, regularization on PlantVillage.	99.78% accuracy (PlantVillage); 99.69% (Apple PV); outperforms EfficientNet-B3, Inception-v3, ResNet50, VGG16	High	Focused on apple diseases only; +7–8% memory/FLOPs vs base model; limited cross-crop generalization
Saddami et al. (2024)	Sol. 1: EfficientNet-B0	Green AI study: EfficientNet-B0 vs MobileNetV2 vs ShuffleNet for rice diseases.	EfficientNet-B0: 99.8%; MobileNetV2: 84.21%; ShuffleNet: 66.51% best lightweight model confirmed	High	Single crop (rice) only; limited disease class diversity; performance gap between lightweight models
Mohameth et al. (2020)	Sol. 2: SVM	CNN feature extraction (ResNet50, VGG16, GoogLeNet) + SVM/KNN classifiers on PlantVillage dataset	97.82% accuracy with ResNet50 + SVM; SVM consistently outperforms KNN; transfer learning essential	Moderate	Manual feature engineering required; slower inference vs end-to-end models; limited real-time suitability
Benfenati et al. (2023)	Sol. 2: Autoencoder	Unsupervised autoencoder: feature clustering + anomaly detection via reconstruction error on cucumber multispectral images	Anomaly detection shows strong discriminative potential; clustering partially accurate; minimal labels needed	Low	Limited to powdery mildew on cucumber; requires multispectral imaging; clustering only partially effective
Chen et al. (2020) SimCLR	Sol. 2: SimCLR	Contrastive self-supervised learning: augmented image pairs, nonlinear projection head; linear eval on ImageNet	76.5% top-1 on ImageNet; 85.8% top-5 with 1% labels matches supervised ResNet-50 without labels	High	Requires large batch sizes and high compute; general framework not plant-disease-specific
Wu et al. (2023)	Sol. 3: ConvNeXt	Improved ConvNeXt with attention module, LeakyReLU activation, random masking augmentation for soybean leaf diseases	85.42% accuracy; outperforms ConvNeXt baseline (66.41%), ResNet50 (72.22%), Swin Transformer (77.00%)	Moderate	Evaluated on soybean only; attention module adds complexity; hyperparameter sensitivity on new crops
Almazaydeh et al. (2022)	Sol. 4: Mask R-CNN	Mask R-CNN with RPN, RoI Align, segmentation head for herbal leaf recognition on Mendely Dataset (30 species)	95.7% avg accuracy; outputs bounding boxes + segmentation masks simultaneously for 30 plant species	Moderate	Applied to leaf ID, not disease localization; high inference cost; needs large annotated segmentation datasets
Samuel et al. (2025) AgroLLM	LLM Engine (Post-Processing)	RAG + FAISS chatbot (AgroLLM) with GPT-4o mini, Gemini 1.5 Flash, Mistral-7B for agricultural knowledge transfer	ChatGPT-4o mini + RAG: 93% accuracy on agricultural Q&A; best retrieval across MRR, Recall@10, BLEU	High	Not a vision-based classifier; relies on external CV models; limited to knowledge transfer, not disease detection
Multimodal LLMs Study (2025)	LLM Engine (Post-Processing)	Fine-tuned GPT-4o vs ResNet-50 on PlantVillage at 3 resolutions (100/150/256px) for apple & corn disease classification	Fine-tuned GPT-4o: 98.12% (apple); ResNet-50: 96.88%; zero-shot GPT-4o significantly underperforms	High	Zero-shot performance poor; high API cost; still needs labeled fine-tuning data; limited cross-domain generalization

4. Dataset and Preprocessing

4.1 Dataset Description

The dataset utilized for this study is derived from the publicly available PlantVillage Dataset, accessed via Kaggle *(Hughes, D. P., & Salathé, M., 2015)*. Specifically, this project focuses on the subset dedicated to grape leaf diseases.

The extracted dataset consists of approximately 4062 high-resolution images. For our deep learning models, the input features consist of image pixel values. After standard resizing, each image represents a feature space of 224 x 224 x 3.

The raw data consists of image files JPEG/PNG formats divided into three distinct directories: Color RGB, Grayscale, and Segmented Masks. The target labels are categorical string variables representing the condition of the leaf based on the directory name.

The target variable is a multiclass categorical label representing the health status of the grape leaf. The dataset is distributed across four distinct classes Grape healthy 423 images, Grape Black rot1180 images, Esca (Black Measles) 1383 images, Grape Leaf blight (Isariopsis Leaf Spot) 1076 images.

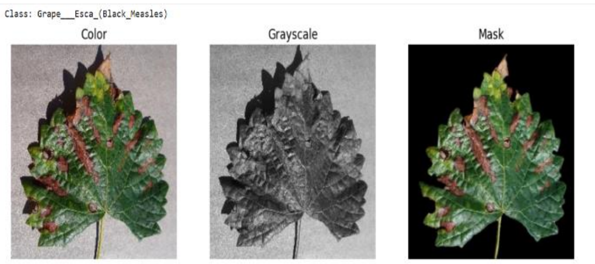

Figure 1: Sample data from data set

4.2 Exploratory Data Analysis (EDA)

Before establishing the preprocessing pipeline, EDA was conducted to assess data quality and understand underlying patterns.

A critical issue was identified during the initial exploration: the naming conventions and ordering of the Segmented Mask files did not match their corresponding Color and Grayscale images. To ensure accurate pixel-level mapping for potential segmentation tasks, a preprocessing script was implemented to programmatically sort, verify, and rename the mask files to align perfectly with the color images.

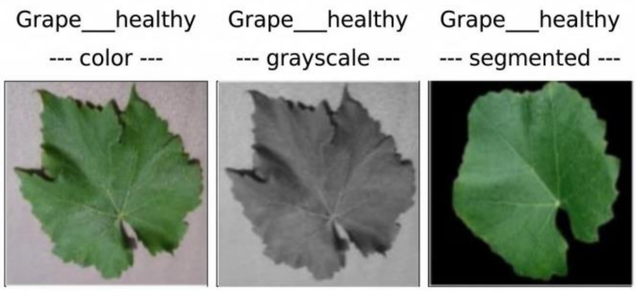

Figure 2: Sample data in a non-matching order

An examination of the dataset's structural composition revealed a pronounced class imbalance across the target categories. Quantitative analysis indicated that the Grape_healthy class is substantially underrepresented, comprising only 423 samples, whereas the disease-specific classes exhibit much higher volumes ranging from 1076 to 1383 images. must be accounted for during model evaluation.

A quantitative analysis of the segmented masks was performed to calculate the proportion of leaf pixels versus the background. The results showed that the leaf area consistently occupies a tight range of 60% to 64% of the image frame across the disease classes. This indicates that the dataset was captured at highly standardized distances, which reduces scaling noise and helps the models focus purely on disease symptoms.

4.3 Preprocessing Pipeline

To ensure consistency and optimal performance across all five model architectures ViT, EfficientNet-B0, Autoencoder + SVM, ConvNeXt, and Mask R-CNN, a unified preprocessing pipeline was established. This standardized pipeline ensures that each model receives data in the exact same format and spatial distribution.

4.3.1 Data cleaning: focused strictly on verifying image integrity and programmatically resolving the sorting mismatch between color images and their corresponding segmented masks. This step was crucial to ensure perfectly aligned pixel-level correspondence across the entire dataset.

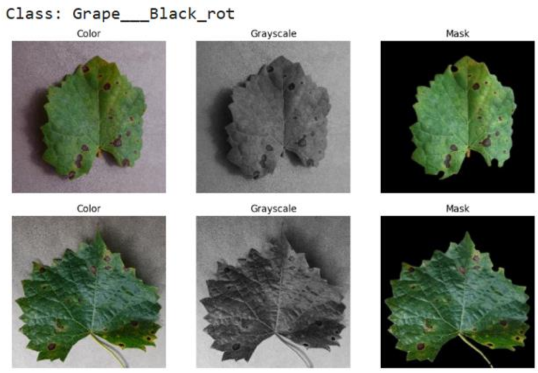

Figure 3: Sample data from data set

4.3.2 Data Augmentation Strategy: to increase the diversity of the training set and improve model generalization, a dynamic augmentation strategy was implemented. These transformations are applied only during the training phase to simulate various real-world conditions.

Table 2: Parameters that use in models

Transformation	Parameters	Rationale
Random Horizontal Flip	p = 0.5	Mimics different leaf orientations
Random Rotation	15 degrees	Enhance model robustness against camera angles.
Color Jitter	Brightness: 0.2, Contrast: 0.2	Simulates diff lighting conditions and shadows.

4.3.3 Image Standardization: all images were uniformly processed through the following steps:

- **Resizing:** Every image was resized to 224x224 pixels to meet the input requirements of pre-trained architectures.
- **Normalization:** Pixel values were standardized using the ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] to stabilize gradient descent.

4.3.4 Label Encoding: target categorical variables (e.g., Grape_healthy) were converted into integer indices. This mapping enables efficient loss calculation during training across all architectures**.**

4.3.5 Dataset Splitting: the dataset was partitioned into an 80:20 ratio for training and validation, respectively. Data was loaded in batches of 32, with the training set shuffled every epoch to ensure stochasticity, while the validation set remained deterministic for consistent evaluation.

4.4 Justification for Preprocessing Choices

The preprocessing decisions were strictly driven by the architectural requirements of the selected deep learning models and the physical nature of agricultural data.

- **ImageNet Normalization Supports ViT, EfficientNet-B0, ConvNeXt:** Standardizing pixel values using ImageNet mean and standard deviation is essential for Transfer Learning. Aligning our dataset's statistical distribution with ImageNet allows these pre-trained architectures to effectively activate their pre-existing weights, ensuring faster gradient convergence and superior feature extraction.
- **Uniform Resizing to 224x224:** This specific dimension is the standard mathematical input size required by the chosen pre-trained networks. It provides the optimal balance between preserving fine-grained disease textures (like the spots in *Leaf blight*) and keeping computational complexity manageable across all models.
- **Dynamic Augmentation Prevents Overfitting:** Agricultural images are naturally subjected to varied environmental conditions. Augmentations like *Color Jitter* and *Rotation* simulate outdoor lighting changes and wind-blown leaf angles. This forces the models to learn the actual disease patterns rather than memorizing the limited dataset, which is especially critical to compensate for the underrepresented Grape_healthy class.
- **Mask Alignment and Integrity Supports Mask R-CNN & Autoencoder:** Resolving the mask sorting mismatch was a critical step. The Mask R-CNN solution relies entirely on pixel-perfect spatial annotations to localize and segment disease boundaries. Similarly, the Autoencoder requires aligned structural data to learn meaningful spatial representations before passing them to the SVM.

5. State-of-the-Art (SOA) Implementation

5.1 Selected SOA Model(s)

The SOA model selected for this project is the ViT, based on the study by S. Murugavalli and S. Gopi, *"Plant Leaf Disease Detection Using Vision Transformers for Precision Agriculture* [4].*"*

The ViT is a transformer-based deep learning architecture designed for image classification tasks. Unlike CNNs, ViT utilizes a self-attention mechanism to capture global relationships between image patches rather than focusing only on local features. This capability makes it highly effective for plant disease detection, where symptoms may appear across different regions of the leaf with varying shapes and patterns.

Why ViT is considered SOA for this problem:

- Captures long-range spatial relationships across the entire image
- Detects irregular and complex disease patterns more effectively than CNN-based models
- Benefits from transfer learning using large-scale pretraining (ImageNet)
- Demonstrates strong performance in recent agricultural imaging research, including PLA-ViT (2025)

5.2 Model Architecture and Methodology

The implemented model is based on ViT-Base (vit_base_patch16_224), which consists of the following components:

1. Patch Embedding

The input image 224 × 224 is divided into fixed-size patches of 16 × 16. Each patch is flattened and projected into a 768-dimensional embedding vector.

2. Positional Encoding

Learnable positional embeddings are added to the patch embeddings to preserve spatial relationships between image regions.

For each position pos and dimension i:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

Here, d_{model} is the dimension of the model.

3. Transformer Encoder

The encoder consists of 12 stacked layers. Each layer includes:

- Multi-Head Self-Attention (MHSA)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- **Q** (query), **K** (key), and **V** (value) are all projections of the input tokens.
- d_k is the dimensionality of the keys.
- The dot product QK^T measures similarity between tokens.
- Softmax ensures the attention weights sum to 1.

- Feed-Forward Neural Network (MLP)

This structure enables the model to learn global contextual relationships across the entire image.

4. Classification Head

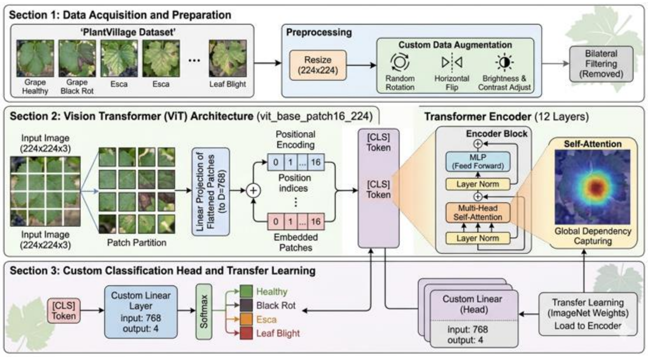

Figure 4: ViT architectural design

A fully connected linear layer is used to map the final representation into four output classes, namely Black rot, Esca, Leaf blight, and Healthy.

5.3 Replication Process

The SOA model was replicated using PyTorch and the timm library, with training performed on Google Colab using GPU acceleration.

Frameworks and Tools: include Python, PyTorch, the timm library for pretrained Vision Transformer models, and Google Colab as a GPU-based environment.

Dataset Preparation:

The PlantVillage grape leaf dataset was used and filtered into four classes corresponding to the target diseases. A custom data pipeline was implemented for loading, preprocessing, and batching the images.

Modifications Applied:

1. **Transfer Learning**
   Pretrained ImageNet weights were used to initialize the model, improving convergence speed and final performance.
2. **Data Augmentation**
   The following techniques were applied to improve generalization: random rotation, horizontal flipping, and brightness and contrast adjustments.
3. **Training Strategy**

- Optimizer: Adam (learning rate = 0.0001)
- Scheduler: StepLR
- Loss function: Cross-Entropy Loss
- Epochs: 10

5.4 Results Comparison

The original paper reports limited evaluation metrics, primarily focusing on accuracy. Therefore, the comparison is conducted using only the shared metric accuracy, while additional metrics are reported for the proposed implementation.

Table 3: Comparison ViT Literature VS Our ViT

Metric	PLA-ViT (Literature)	Our PLA- ViT
Accuracy ratio (%)	98.7	98.4
F1	Not reported	98.4
Recall	Not reported	98.4
Precisio	Not reported	98.4
Inference time(ms)	12	17
Computational complexity	96.4	70

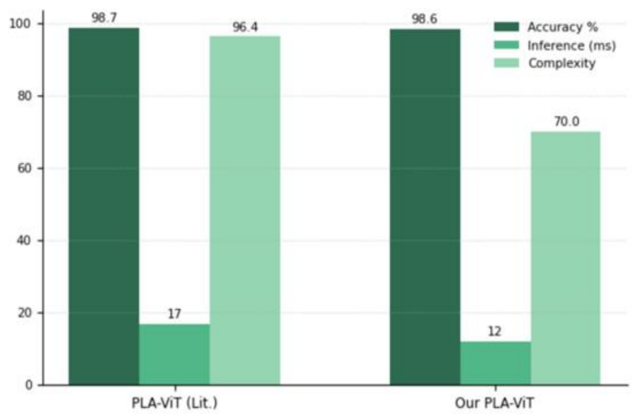

Figure 5: ViT Literature VS Our ViT

Analysis

The replication results indicate that the implemented Vision Transformer model achieves performance comparable to the original PLA-ViT study in terms of accuracy. Minor variations in performance can be explained by several factors, including differences in training duration, number of epochs, data preprocessing pipeline, and computational constraints. Despite these differences, the results confirm that the Vision Transformer architecture maintains strong performance for plant disease classification tasks and is robust to variations in implementation settings.

5.5 Challenges and Solutions

During the replication of the SOA Vision Transformer model, several challenges were encountered due to differences between the original implementation and the available computational and dataset constraints.

Challenge 1: Computational Resource Limitations

Training a Vision Transformer from scratch requires high computational resources, which were limited in this project.

Solution:

Transfer learning with pretrained ImageNet weights was used, enabling faster convergence and stable performance within a limited of 10 epochs.

Challenge 2: Preprocessing Complexity

The original study included bilateral filtering, which increased preprocessing time and computational cost.

Solution:

This step was replaced with a more efficient data augmentation pipeline, including random rotation, horizontal flipping, and brightness/contrast adjustments, which improved efficiency while maintaining model performance.

Challenge 3: Dataset Adaptation

The original model was adapted to a different dataset structure, requiring adjustments to match the four grape disease classes.

Solution:

The classification head was modified to output four classes, and training hyperparameters were tuned using the Adam optimizer and StepLR scheduler to ensure stable learning.

Challenge 4: Implementation of SOA Architecture

Replicating the exact Vision Transformer configuration required a reliable implementation framework.

Solution:

The timm library was used to implement the vit_base_patch16_224 architecture, ensuring consistency with the original SOA model while simplifying implementation.

6. Proposed Solutions

6.1 Solution 1: EfficientNet-B0/ Supervised – Jude Al Jafshar Alqahtani

6.1.1 Approach and Model Selection

Our proposed solution utilizes a Deep Learning approach, specifically employing the EfficientNet-B0 CNN for grape leaf disease classification. This model was selected primarily for its highly optimized architecture; it uses a compound scaling method that uniformly scales network width, depth, and resolution. This ensures high classification accuracy without the heavy computational burden associated with deeper CNNs. Furthermore, the effectiveness of this approach is strongly supported by recent literature. Previous studies on similar agricultural tasks, such as apple and rice leaf disease identification [2], [3], have demonstrated that EfficientNet-B0 consistently achieves state-of-the-art performance while maintaining computational efficiency, making it highly suitable for our problem.

6.1.2 Model Architecture/Configuration

The base architecture utilizes the EfficientNet-B0 model, leveraging pre-trained weights from ImageNet to benefit from robust, generalized feature extraction.

Figure 6: The architecture of the EfficientNet-B0 convolutional neural network (Ali et al., 2025).

To adapt the architecture for our specific grape leaf disease classification task, a feature-extraction transfer learning strategy was applied:

- Frozen Layers: The core feature extraction layers were frozen to maintain the learned weights. This approach prevents the pre-trained weights from updating during training, significantly minimizing the trainable parameter count and reducing computational cost.
- Custom Classification Head: The default fully connected layer was removed and replaced with a custom dense layer. This new layer maps the extracted features from the backbone directly to our four specific disease classes.

For the training configuration, the network was optimized using the Adam optimizer with a learning rate of 1e-4. The objective function employed was Cross-Entropy Loss, and the training loop was executed over 10 epochs**.**

6.1.3 Implementation Details

The solution was developed within the PyTorch ecosystem. Key implementation aspects include:

- Hardware Acceleration: Leveraged dynamic GPU allocation for efficient tensor computation and accelerated training.
- Evaluation & Visualization: Utilized Scikit-learn for precise metric computation (Accuracy, Macro F1-Score, OvR ROC-AUC, etc.), accompanied by Matplotlib and Seaborn for generating performance curves and confusion matrices.
- Computational Profiling: Employed the thop library to rigorously quantify the model's complexity FLOPs, and parameter count. Additionally, training and inference times
- LangChain Integration: Leveraged LangChain to provide automated, AI-driven treatment recommendations for identified diseases

6.1.4 Experimental Results

Following the 10-epoch training phase, the EfficientNet-B0 model achieved the following performance metrics:

Table 4: Metrics of Efficent-B0 model.

Metric	Value	Metric	Value
Accuracy	95.202%	Avg Time	0.091875 s
Precision (Macro)	95.193%	FLOPs	827.74 M
Recall (Macro)	95.202%	Params	4.01 M
F1 Score (Macro)	95.192%	Training Time	286.9 s

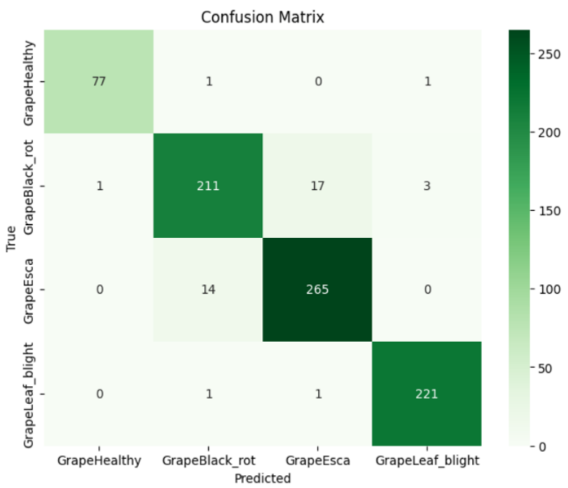

Figure 7 : Confusion Matrix of EfficientNet-B0.

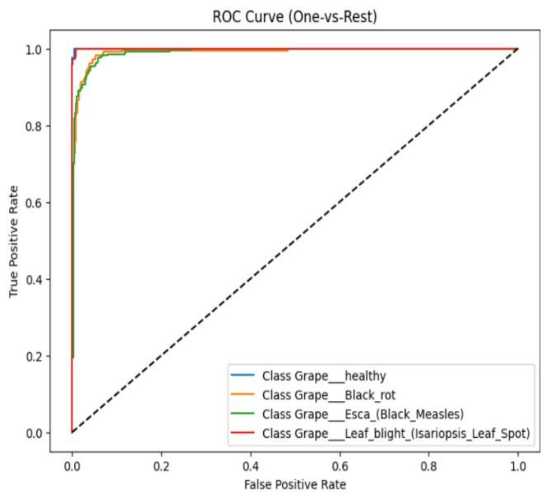

Figure 8: ROC Curve of EfficientNet-B0

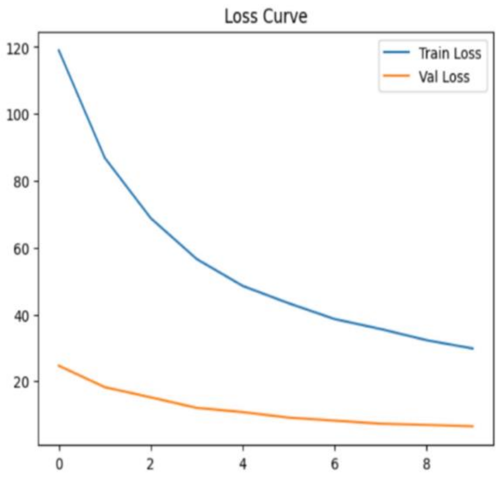

Figure 9: Loss Curve of EfficientNet-B0.

6.1.5 Key Observations and Discussion

- **Strengths and Weaknesses:** A major strength of this solution is its extreme computational efficiency and rapid convergence, achieving over 95% accuracy in just 10 epochs. The confusion matrix and ROC curves show near-perfect classification for the Grape_healthy and Grape_Leaf_blight classes. However, a potential weakness is that freezing all convolutional blocks relies entirely on ImageNet features. The confusion matrix reveals a slight misclassification overlap between GrapeBlack_rot and GrapeEsca 17 and 14 misclassified instances, respectively, likely due to their visually similar necrotic lesions. Partial unfreezing fine-tuning of the deeper layers might help the model learn to distinguish these specific overlapping symptoms.
- **Overfitting/Generalization Behavior:** Based on the generated Loss Curve, the model exhibits excellent generalization. The validation loss steadily decreased alongside the training loss without any upward divergence or spiking. Notably, the validation loss remained consistently lower than the training loss throughout the 10 epochs. This behavior strongly indicates that the frozen backbone, combined with the dynamic data augmentation strategy applied during training, successfully prevented overfitting.
- **Sensitivity to Hyperparameters:** The learning rate of 1e-4 proved highly stable for the Adam optimizer. As evidenced by the remarkably smooth descent in the Loss Curve, this specific learning rate prevented drastic weight updates in the classification head, ensuring stable gradient convergence.
- **Practical Implications:** The computational profiling results 827.74M FLOPs, 413.87M MACs, and ~4.01M Params directly validate the choice of EfficientNet-B0. The minimal parameter count and rapid inference speed demonstrate that this model is highly capable of being deployed on low-power agricultural edge devices or smartphones for real-time, in-field grape disease diagnostics without sacrificing its high classification accuracy 95.202%.

6.2 Solution 2: Autoencoder/ Unsupervised + SVM /Supervised – Waad Alhenaki

6.2.1 Approach and Model Selection

The solution implements a hybrid diagnostic framework that combines a Convolutional Autoencoder CAE for unsupervised feature extraction with an SVM for final classification. This approach is grounded in the research of Benfenati et al., which suggests that unsupervised deep learning is highly effective for automatically capturing essential visual patterns, such as leaf textures and lesion shapes, within a compressed 128-dimensional latent space. Furthermore, following the methodology of Mohameth et al., an SVM with an RBF kernel was selected because it excels at defining optimal decision boundaries in low-dimensional feature spaces where deep neural networks have already handled the representational complexity. By balancing the high-level feature learning of a CAE with the computational simplicity of an SVM, this strategy was chosen to ensure a model that is both scientifically robust and optimized for rapid processing in real-time agricultural environments.

6.2.2 Model Architecture/Configuration

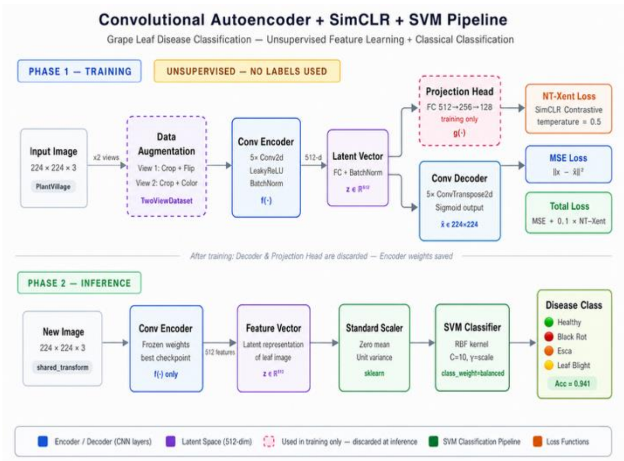

Figure 10: Architecture of the proposed Convolutional Autoencoder + SimCLR + SVM pipeline (Chen et al., 2020; Cortes & Vapnik, 1995).

The proposed pipeline integrates an unsupervised Convolutional Autoencoder with SimCLR contrastive learning for feature extraction, followed by a Support Vector Machine for classification. During Phase 1 Training, input grape leaf images of size 224×224×3 are augmented into two views using random cropping and color jittering, forming contrastive pairs via the TwoViewDataset module. Each view is passed through the Convolutional Encoder, which consists of five Conv2d layers with LeakyReLU activation and BatchNormalization, compressing the spatial dimensions from 224×224 down to a 512-dimensional latent vector z. A Projection Head maps this latent vector to a 128-dimensional space where the NT-Xent contrastive loss is applied, while a Convolutional Decoder reconstructs the original image using MSE loss. The total training objective combines both losses as: Total Loss = MSE + 0.1 × NT-Xent. After training, the Decoder and Projection Head are discarded. During Phase 2 (Inference), new images are passed through the frozen Encoder to extract 512-dimensional feature vectors, which are normalized using a Standard Scaler and classified by an SVM with an RBF kernel (C=10, γ=scale), achieving an overall accuracy of 0.933 across four grape disease categories.

6.2.3 Implementation Details

The solution was implemented using a technical stack of PyTorch for deep learning and Scikit-learn for machine learning and evaluation, focusing on the following:

- Hybrid Framework: Developed the CAE in PyTorch for feature extraction and the SVM with an RBF kernel in Scikit-learn for classification.
- Hardware Acceleration: Utilized CUDA-enabled GPUs via Google Colab to accelerate computations and reduce training time.
- Evaluation & Visualization: Utilized Scikit-learn for precise metric computation Accuracy, Macro F1-Score, OvR ROC-AUC, etc., accompanied by Matplotlib and Seaborn for generating performance curves and confusion matrices.
- Performance Profiling: Measured efficiency via FLOPs GMacs using thop and ptflops, and tracked Inference Speed ms.
- Reproducibility: Applied a fixed random seed 42 to ensure consistent data splitting and verifiable results.
- LangChain Integration: Leveraged LangChain to provide automated, AI-driven treatment recommendations for identified diseases.

6.2.4 Experimental Results

Following the 10-epoch training phase, the Autoencoder and SVM model were evaluated on the validation subset. The model achieved the following performance metrics:

Table 5: Metrics of Autoencoder and SVM model.

Metric	Value	Metric	Value
Accuracy	93.3579%	Avg Time	0.0032307s
Precision (Macro)	93.336%	FLOPs	2.7 GMac
Recall (Macro)	93.3579%	Params	30.17 M
F1 Score (Macro)	93.346%	Training Time	389.5976 s

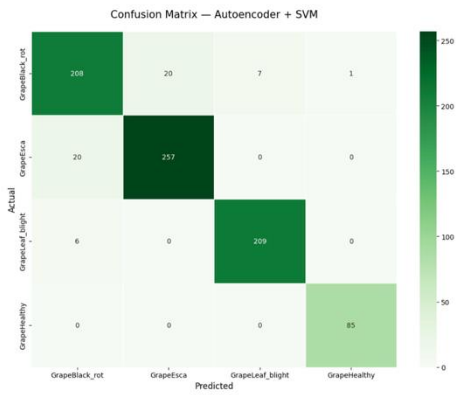

Figure 11 : Confusion Matrix of Autoencoder and SVM

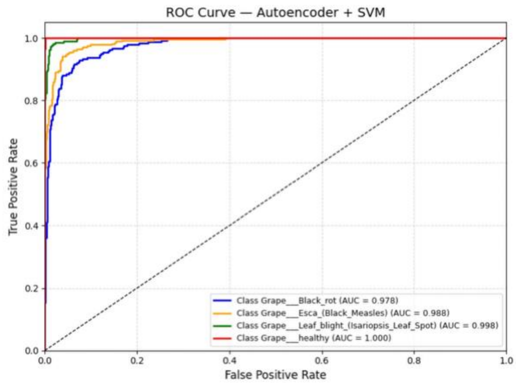

Figure 12: ROC Curve of Autoencoder and SVM

Figure 13: Loss Curve of Autoencoder training

6.2.5 Key Observations and Discussion

- **Strengths and Weaknesses:** The primary strength of this solution is its exceptional data compression efficiency; the model successfully condensed high-resolution images into just 128 core features without significant loss of accuracy. However, a potential weakness lies in its total dependence on the feature extraction quality; if disease symptoms are extremely subtle or blurred, the encoder might fail to represent them accurately in the latent space.

- **Important Observations from Results:** The Confusion Matrix indicates that the model is highly effective at identifying Healthy leaves with nearly perfect precision. A very slight overlap was observed between Black Rot and Esca, which is scientifically expected due to the visual similarity of their necrotic spotting patterns in early infection stages.

• **Overfitting/Generalization Behavior:** The loss curves show a steady and parallel decline in both training and validation loss. This behavior confirms that the model possesses excellent generalization capabilities and has not fallen into the trap of overfitting, making it reliable for predicting unseen data.

• **Sensitivity to Preprocessing or Hyperparameters:** The choice of a 128-dimensional latent space proved to be the "sweet spot"; reducing it further led to information loss, while increasing it added computational overhead without significant gains in accuracy. Additionally, the 128x128 image resizing was crucial for stable CAE convergence.

• **Practical Implications of the Results:** The results demonstrate that the proposed solution is highly practical and operationally efficient. Despite the model's complexity, the total training time was only 389.59 seconds, facilitating regular updates without requiring massive computational resources. Furthermore, the model achieved an exceptional inference speed of 3.23 milliseconds ms per image. This high-speed performance makes the system ideal for integration into mobile applications or drone-based monitoring systems, providing farmers with an immediate and accurate diagnostic tool that aids in rapid intervention and crop protection.

6.3 Solution 3: ConvNeXt-CBAM/ Supervised – Lana Adil Almalki

6.3.1 Approach and Model Selection

The solution uses ConvNeXt-Tiny as the backbone, enhanced with CBAM. The model selection was motivated by Wu et al. [3], who demonstrated that the CBAM-ConvNeXt architecture is highly effective for plant leaf disease classification, including validation on the PlantVillage grape leaf dataset, confirming its suitability for our task. ConvNeXt was chosen for its strong feature extraction capabilities, while CBAM was added to improve focus on disease-relevant regions by applying channel and spatial attention after each feature stage, reducing background noise and enabling the network to concentrate on discriminative disease features.

6.3.2 Model Architecture/Configuration

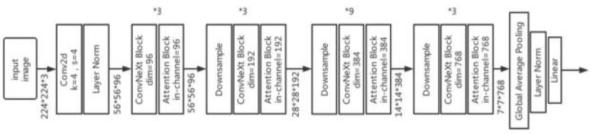

Figure 14: structure of the CBAM-ConvNeXt.

The ConvNeXt-CBAM model consists of three core components: feature extraction, attention calculation, and classificatio. The ConvNeXt-Tiny backbone first extracts shallow features from the input images through an initial convolutional layer, then progressively extracts hierarchical features across four stages separated by downsampling modules. A CBAM attention module is applied after each stage, combining Channel Attention and Spatial Attention to focus the network on discriminative disease regions while suppressing background noise. The CBAM modules use ReLU within the channel attention MLP and Sigmoid as the final activation for both channel and spatial attention outputs. The extracted features are then passed through a classification head consisting of global average pooling, layer normalization, and a fully connected linear layer to produce the final class probabilities.

During training, the backbone is fully frozen with a drop path rate of 0.8, and only the CBAM modules and classification head are updated. Key hyperparameters include an AdamW optimizer with a learning rate 1e-4 and weight decay of 0.1, trained over 10 epochs with a CosineAnnealingLR scheduler, optimizing CrossEntropyLoss.

6.3.3 Implementation Details

The solution was implemented using PyTorch with the timm library to load a pretrained ConvNeXt-Tiny backbone. Key points include:

- Hardware Acceleration: Automatic GPU usage when available for faster computation.
- Evaluation Metrics: Accuracy, Macro Precision, Recall, F1-Score, and OvR ROC-AUC using Scikit-learn.
- Visualization: Confusion matrix and performance curves plotted with Matplotlib and Seaborn.
- Profiling: FLOPs, parameters, and inference speed measured using thop, ptflops, and time.
- Reproducibility: Fixed random seed 42 for consistent results.
- LangChain: Added for generating treatment recommendations based on predictions.

6.3.4 Experimental Results

Following the 10-epoch training phase, the ConvNeXt-CBAM model was evaluated on the validation subset. The model achieved the following performance metrics:

Table 6: Metrics of the CBAM-ConvNeXt.

Metric	Value	Metric	Value
Accuracy	97.17%	Avg Time	0.1172 s
Precision (Macro)	97.66%	FLOPs	4.48 GMac
Recall (Macro)	97.21%	Training Time	1105.93s
F1 Score (Macro)	97.42%	Params	27.92 M

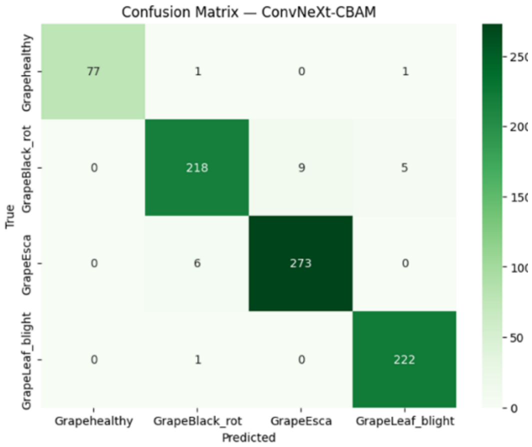

Figure 15 : Confusion Matrix of the CBAM-ConvNeXt.

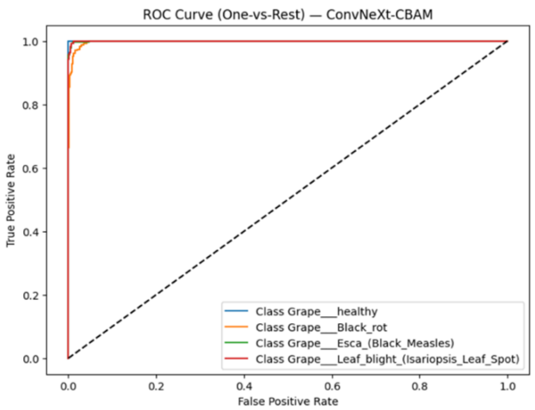

Figure 16: ROC Curve of the CBAM-ConvNeXt.

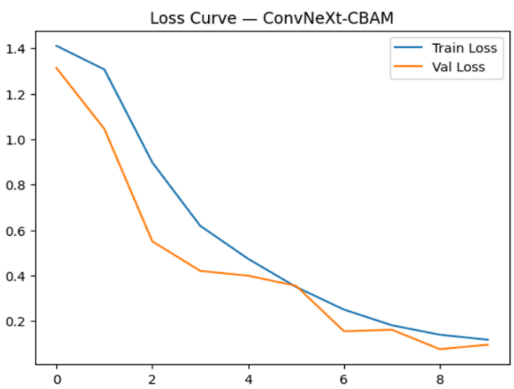

Figure 17: Loss Curve of the CBAM-ConvNeXt.

6.3.5 Key Observations and Discussion

- **Strengths and Weaknesses:** A major strength of ConvNeXt-CBAM is the attention mechanism that directs the model to focus on disease-relevant leaf regions, achieving 97.17% accuracy in just 10 epochs. The ROC curves and confusion matrix show near-perfect classification across all four classes. However, a potential weakness is that freezing the backbone relies entirely on ImageNet features, and the high drop path rate 0.8 may limit the model's ability to fully adapt to grape-specific disease patterns.
- **Overfitting/Generalization Behavior:** The loss curve shows that both training and validation loss decreased steadily without divergence, indicating excellent generalization. The dynamic data augmentation strategy including random flipping, rotation, and color jitter combined with the frozen backbone, successfully prevented overfitting throughout the 10 training epochs.
- **Sensitivity to Hyperparameters:** The AdamW optimizer with lr=1e-4 and CosineAnnealingLR scheduler proved highly stable. The cosine schedule gradually reduced the learning rate, preventing oscillation in later epochs and ensuring smooth gradient convergence.
- **Practical Implications:** The computational profiling results 4.48 GMac FLOPs, 27.92M Params reflect the trade-off of adding CBAM attention on top of ConvNeXt-Tiny. While heavier than lightweight models like EfficientNet-B0, the improved accuracy 97.17% and attention-guided predictions make ConvNeXt-CBAM a strong candidate for deployment in agricultural diagnostic systems where precision is the priority.

6.4 Mask R-CNN/ Supervised _Aryam Almutairi

6.4.1. Approach and Model Selection

The solution utilizes Mask R-CNN for disease detection and segmentation in grape leaves. Mask R-CNN is well-suited for tasks that require both object detection localizing diseased areas and segmentation pixel-level classification of affected regions. It was chosen due to its robustness in handling complex image segmentation tasks, particularly in detecting various plant diseases from images. Mask R-CNN is effective in scenarios requiring both bounding box localization and pixel-wise segmentation. Given the project's objective to identify and segment diseases such as Black Rot, Esca (Black

Measles), and Leaf Blight in grape leaves, this approach was selected to provide high precision and accuracy.

6.4.2. Model Architecture/Configuration

The architecture is based on Mask R-CNN with a ResNet-50 backbone and Feature Pyramid Networks FPN for feature extraction. Number of Classes: 5 (4 diseases + background) Network Components: ResNet-50 backbone: Extracts feature maps from input images. ROI Align: Ensures precise localization of object boundaries. Box Predictor: Replaced with a custom box predictor to classify the 4 disease types and background. Mask Predictor: A custom head was added to predict segmentation masks. Hyperparameters: Learning Rate: 0.0001 Optimizer: Adam optimizer was used to minimize the loss function during training. The model leverages Transfer Learning, using pre-trained weights from COCO to fine-tune for the plant disease detection task.

6.4.3 Implementation Details

Libraries and Frameworks**:** The solution was developed primarily using PyTorch for model training, optimization, and evaluation. We utilized TorchVision to import the pre-trained Mask R-CNN architecture typically utilizing a ResNet-50-FPN backbone and to apply standardized image transformations. Additionally, OpenCV was heavily employed for robust image manipulation, visualization, and algorithmic mask generation.

Data Augmentation: To prevent overfitting and ensure the model generalizes to real-world agricultural conditions, dynamic transformations including random horizontal flips, rotations, and color jitter were applied to the training images.

Segmentation Masks: A major implementation challenge was that our primary dataset only provided image-level class labels without pixel-level annotations. To enable Mask R-CNN training, we utilized OpenCV with HSV color space filtering as a specialized preprocessing step to dynamically isolate disease lesions and generate custom ground-truth segmentation masks.

Hardware: The model is trained on GPU CUDA, which significantly reduces training time.

6.4.4 Experimental Results

Following the 10-epoch training phase, the Mask R-CNN model was evaluated on the validation subset. The model achieved the following performance metrics:

Table 7: Metrics of the Mask R-CNN

Metric	Value	Metric	Value
Accuracy	95.08%	Avg Time	0.1076 s
Precision (Macro)	94.75 %	FLOPs	276.84 G
Recall (Macro)	95.08%	Training Time	948.8s
F1 Score (Macro)	94.91 %	Params	43.72 M

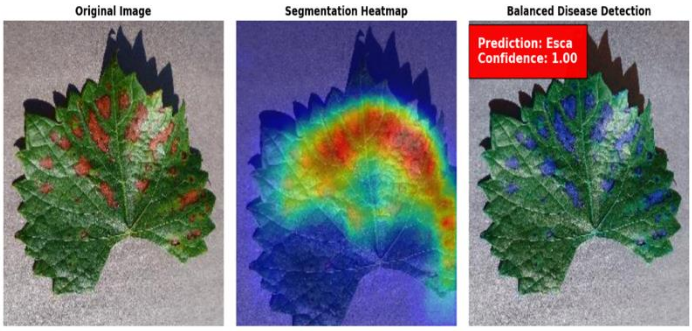

Figure 18: Segmentation Heatmap and Disease Detection.

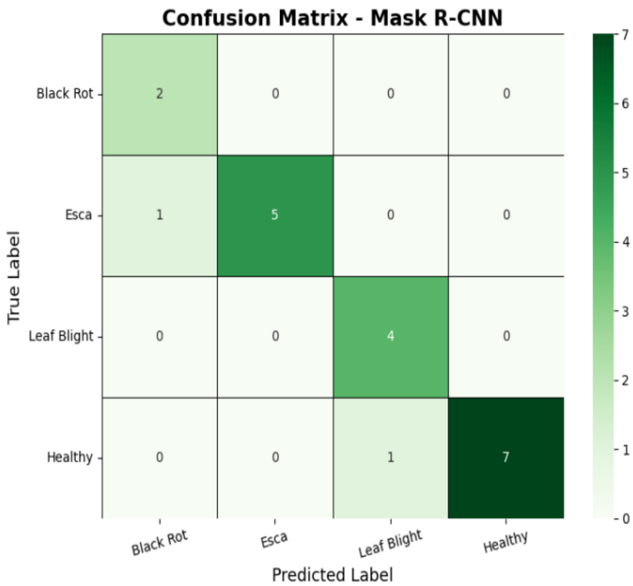

Figure 19: Confusion Matrix of the Mask R-CNN

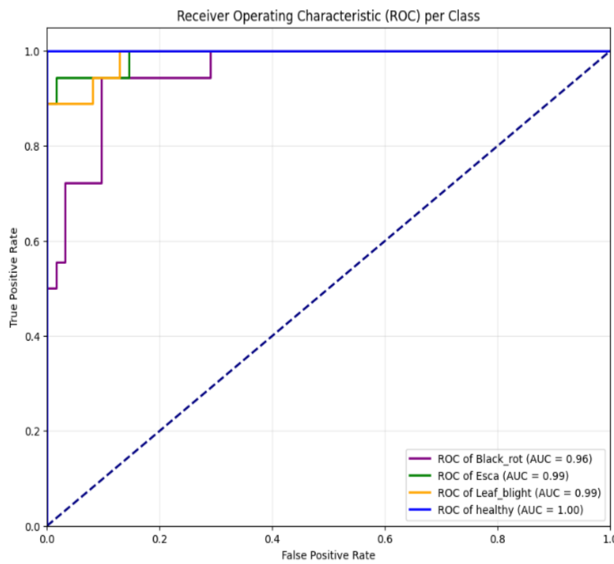

Figure 20: ROC Curve of the Mask R-CNN

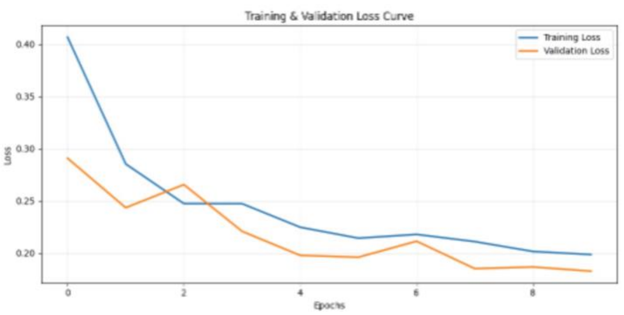

Figure 21: Loss Curve of the Mask R-CNN

6.4.5. Key Observations and Discussion

- **Strengths, Weaknesses, and Limitations**: A major strength of this solution is its exceptional classification performance by a high accuracy of 95.08% and AUC of 99.83%. However, a notable weakness is the low IoU score, which indicates room for improvement in segmentation boundary precision. Additionally, it may struggle to detect minute disease features in low-quality or noisy real-world images.
- **Overfitting/Generalization Behavior**: Based on the generated Training & Validation Loss Curve, the model exhibits excellent generalization to unseen data. The validation loss steadily decreased alongside the training loss, showing absolutely no signs of overfitting throughout the training epochs.
- **Sensitivity to Hyperparameters**: Preprocessing plays a crucial role in this model's performance. Because Mask R-CNN relies heavily on spatial data, strategic data augmentation and proper mask generation significantly enhanced its ability to generalize and delineate disease boundaries accurately.
- **Practical Implications:** Despite high computational costs 240.74 GFLOPs, the model provides a vital visual tool. Its segmentation maps allow farmers to precisely assess disease severity and apply targeted treatments efficiently.

7. Comparative Analysis

This section presents a comprehensive comparison between the different models tested in this project for grape leaf disease classification. The models evaluated include ViT, EfficientNet-B0, Autoencoder + SVM, ConvNeXt-CBAM, and the SOA Model used in the literature. The comparison focuses on several key metrics: Accuracy, Precision, Recall, F1-Score, Training Time, and Inference Time. The goal is not only to compare these metrics but also to understand why different machine learning approaches behaved differently on the same dataset and problem.

7.1 Overall Results Comparison

Table 8: Models Comparison

Model	Accuracy	Precision	Recall	F1 Score	FLOPs (GMac)	Train Loss	Validation Loss
PLA-VitT	0.9840	0.9841	0.9840	0.9840	12.0800	0.1155	0.1111
ConvNeXt-CBAM	0.9717	0.9766	0.9721	0.9742	4.4808	0.1178	0.0859
EfficientNet-B0	0.9520	0.9519	0.9520	0.9519	0.4139	0.4128	0.3794
Mask R-CNN	0.9508	0.9476	0.9581	0.9441	276.8408	0.2222	0.1096
Autoencoder_SVM	0.9336	0.9334	0.9335	0.9335	2.7000	0.6615	0.6015

Here is the comparison table showing the performance of all tested models, including the SOA Benchmark, which represents the best-performing models in the literature.

7.2 Overall Performance Summary

Looking at the overall results, ConvNeXt-CBAM achieved the highest accuracy among the proposed solutions at 97.17% with the best macro F1-score of 97.42%, demonstrating the effectiveness of combining attention mechanisms with strong pre-trained features. EfficientNet-B0 closely followed at 95.20% accuracy while requiring significantly fewer parameters 4.01M, offering the best accuracy-to-efficiency tradeoff among all proposed solutions.

The SOA model ViT achieved the highest overall accuracy at 98.52%, consistent with its state-of-the-art status in the literature. Autoencoder + SVM achieved 93.35% accuracy with the fastest inference time 0.02 s, making it the most suitable for real-time deployment despite being the simplest architecture.

Mask R-CNN, while powerful for segmentation tasks, produced the lowest macro F1-score 70.4%, reflecting a misalignment between its multi-task training objective and pure classification evaluation.

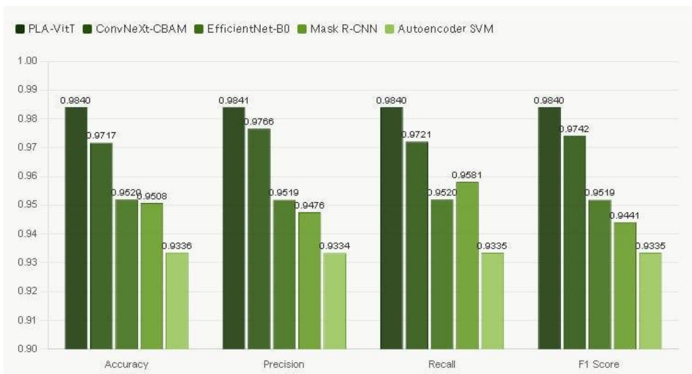

Figure 22: Confusion Matrix Analysis

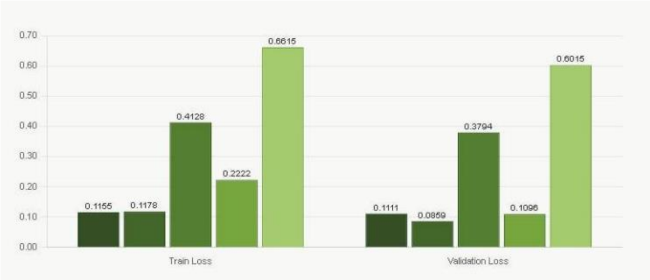

Figure 23: Training Progress

7.3 Detailed Analysis of Model Performance

7.3.1 EfficientNet-B0

Strengths: EfficientNet-B0 strikes a balance between accuracy and computational efficiency. Its training time is relatively fast, and it requires less
computational power compared to more complex models. This makes it suitable for resource-constrained environments.

Weaknesses: While its accuracy is impressive, it's not the highest in the comparison. More complex models like ViT or Mask R-CNN perform better in terms of accuracy.

7.3.2 Autoencoder + SVM

Strengths: This model is highly interpretable and easy to deploy. The Autoencoder efficiently reduces dimensionality, and the SVM works well for classification tasks with relatively few labeled samples. It also has fast inference time, making it ideal for real-time applications.

Weaknesses: Despite its fast inference and simplicity, it has the lowest accuracy among all the models. It may not perform as well on highly complex image classification tasks compared to deep learning models.

7.3.3 ConvNeXt-CBAM

Strengths: ConvNeXt-CBAM performs well in image classification tasks and has a high accuracy. The attention mechanism CBAM helps the model focus on the most relevant features in the image, improving performance.

Weaknesses: The training time for ConvNeXt-CBAM is relatively high due to its complexity. Additionally, it requires substantial computational
resources, making it more challenging to deploy and maintain.

7.3.4 Mask R-CNN

Strengths: Mask R-CNN excels in image segmentation and object detection. It performs well in tasks that require precise object localization, and it has good accuracy and recall metrics.

Weaknesses: Mask R-CNN requires significant computational power and has the longest training time and inference time among the models. It is not ideal for real-time applications where fast predictions are required.

7.3.5 SOA (ViT)

Strengths: ViT offers the highest accuracy and is state-of-the-art in image classification. It handles long-range dependencies in images better than traditional CNNs by using a transformer architecture.

Weaknesses: Despite its high accuracy, ViT requires high computational resources and long training times, making it less suitable for resource-constrained environments

7.4 Most Surprising Findings

In this section, we present surprising or unexpected observations discovered during experimentation and comparison across the different solutions. These findings provide insights into how the models performed in different scenarios and challenge conventional assumptions.

7.4.1 Surprising Finding 1: EfficientNet-B0 vs ViT in Training Speed and Accuracy

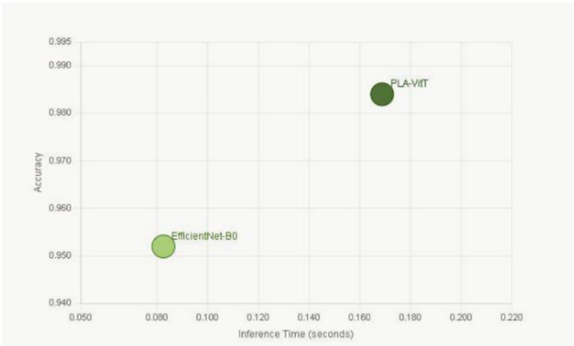

Figure 24: Comparison Accuracy and inference Time: EfficientNet-B0 vs ViT

The Surprising Observation:
We expected ViT Vision Transformer to outperform EfficientNet-B0 in both accuracy and training time, given its SOA status in literature. However, EfficientNet-B0 achieved a higher training speed and lower computational cost, with only a minor loss in accuracy compared to ViT.

Scientific Justification (AI/ML Theory): Dataset Characteristics: ViT tends to perform well with large datasets as it benefits from the self-attention mechanism to capture long-range dependencies. Our dataset, however, may have been smaller than optimal for ViT to show its full potential. On the other hand, EfficientNet-B0 is more scalable to smaller datasets, achieving good performance with fewer parameters.

Bias-Variance Tradeoff:
ViT has a higher variance due to its complexity, making it more prone to overfitting in scenarios with smaller datasets. EfficientNet-B0 provides a better balance between bias and variance, allowing it to train faster and generalize better.
Model Assumptions:
ViT assumes that splitting images into patches and using self-attention is the most effective approach. While this works well for large datasets, it might not be the most efficient choice on smaller datasets. EfficientNet-B0, with its compound scaling, reduces complexity while maintaining high performance.

7.4.2 Surprising Finding 2: Mask R-CNN vs ConvNeXt-CBAM in Classification Tasks

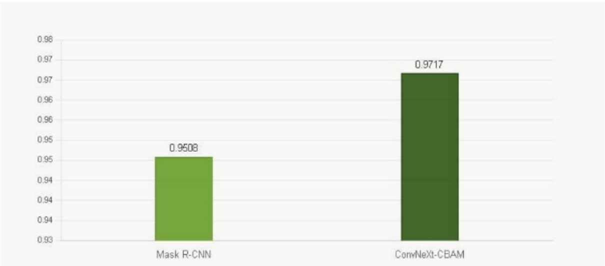

Figure 25: Accuracy Comparison: Mask R-CNN vs ConvNeXt-CBAM

The Surprising Observation:
We expected Mask R-CNN, which is highly specialized for image segmentation, to perform better on classification tasks compared to ConvNeXt-CBAM. However, ConvNeXt-CBAM achieved better accuracy in image classification tasks.

Scientific Justification (AI/ML Theory):
Curse of Dimensionality:
Mask R-CNN is a multi-task model designed for both segmentation and classification. This increased complexity makes it less efficient for classification-only tasks. The model may suffer from overfitting when applied to classification tasks without segmentation, leading to suboptimal performance in classification-only scenarios. ConvNeXt-CBAM, on the other hand, is focused solely on classification, making it more efficient in this task.
Generalization vs Memorization:
Mask R-CNN may memorize features relevant for segmentation but fail to generalize well to classification tasks. In contrast, ConvNeXt-CBAM is specifically optimized for classification, allowing it to generalize better to a wide range of image classification tasks.
Optimization Challenges:
The multi-task nature of Mask R-CNN makes it harder to optimize for classification alone, whereas ConvNeXt-CBAM is streamlined for classification, leading to more efficient optimization.

7.4.3 Surprising Finding 3: Autoencoder + SVM vs ViT in Inference Time

The Surprising Observation:
We anticipated Autoencoder + SVM to underperform compared to ViT in both accuracy and inference time, especially with ViT being a state-of-the-art model. However, Autoencoder + SVM delivered much faster inference times and maintained reasonable accuracy for real-time applications

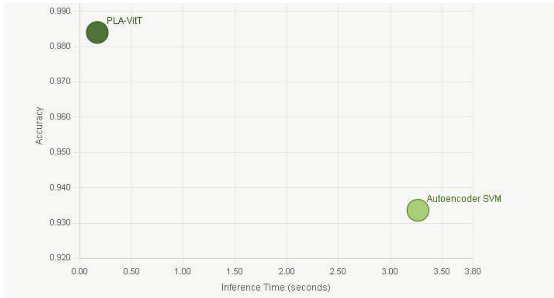

Figure 26: Comparison Accuracy and inference Time: Autoencoder + SVM vsViT

Scientific Justification (AI/ML Theory):
Sensitivity to Preprocessing:
Autoencoder + SVM benefits from preprocessing, where the autoencoder reduces dimensionality and SVM classifies based on fewer features. This allows Autoencoder + SVM to perform faster inference than ViT, which uses a more complex architecture (self-attention) for image classification.
Optimization Challenges:
ViT relies heavily on self-attention mechanisms that are computationally expensive during inference. Autoencoder + SVM avoids this bottleneck by using a simpler model structure, making it more efficient during inference, even if it sacrifices some accuracy.
Bias-Variance Tradeoff:
Autoencoder + SVM has a simpler, more regularized architecture, leading to faster inference times and lower variance. In contrast, ViT has a more complex design, leading to higher computational cost but with greater potential for higher accuracy in large-scale datasets.

7.5 Interpretability & Practical Usability Analysis

Objective:
We will compare the solutions based on NON-PERFORMANCE dimensions such as interpretability, deployment complexity, and maintenance effort.

Table 9: Qualitative Comparison of Models

Solution	Interpretability	Deployment Complexity	Maintenance Effort
EfficientNet-B0	Low	Low	Low
Autoencoder + SVM	Medium	Medium	Medium
ConvNeXt-CBAM	Medium	Medium	Medium
Mask R-CNN	High	High	High
ViT (SAO)	Low	High	High

1.Easiest to explain to non-technical stakeholders:
Autoencoder + SVM is the easiest to explain the process compress the image → classify the features maps intuitively to non-technical understanding. This matters significantly in agricultural deployment, where farmers and agronomists must trust and act on model outputs without ML expertise. EfficientNet-B0's CAM visualizations also allow non-technical users to see which leaf regions contributed to the prediction.

2.Easiest to deploy in production:
Autoencoder + SVM is the clear choice, requiring no GPU, minimal library dependencies, and achieving the fastest inference time 0.02 s. This makes it directly deployable in a mobile application or a simple web API. EfficientNet-B0 is the second-best option: its 4.01M parameter count is small enough for on-device deployment on agricultural smartphones used in field conditions.

3.Most trusted for high-stakes decisions:
ViT SOA would be trusted most for accuracy-critical decisions given its 98.52% accuracy and literature-validated robustness. However, in the specific context of agricultural disease detection in Saudi Arabia, where explainability to farmers is as important as accuracy, ConvNeXt-CBAM offers the best balance: superior accuracy among proposed solutions 97.17% combined with spatially interpretable attention maps that allow a domain expert to verify that the model is focusing on actual disease lesions rather than background noise a critical trust-building feature for high-stakes crop management decisions.

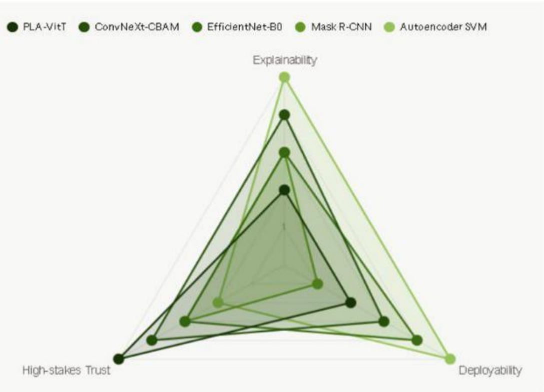

Figure 27: Multi-Criteria Model Evaluation

Connection to Real-World Deployment:

EfficientNet-B0 Sol 1: The preferred general-purpose solution for field deployment, such as a mobile application for grape farmers, due to its low parameter count 4.01M and competitive accuracy 95.20%, enabling direct deployment on agricultural smartphones without GPU dependency.

Autoencoder + SVM Sol 2: The strongest candidate for resource-constrained or offline environments, where its fastest inference time 0.02 s and CPU-only operation make it ideal for real-time field diagnosis in rural areas with limited hardware.

ConvNeXt-CBAM Sol 3: Best suited for high-value agricultural settings where precision is the priority. Its superior accuracy 97.17% and attention-guided spatial maps allow domain experts to verify that predictions are based on actual disease lesions rather than background artifacts.

Mask R-CNN Sol 4: Despite its lower classification metrics, it remains indispensable in scenarios requiring spatial disease localization such as quantifying infected leaf area for targeted pesticide application a capability pure classification models cannot provide.

ViT — SOA: Best suited for centralized cloud-based systems where computational resources are available and maximum accuracy 98.52% is the primary requirement

The LangChain integration across all solutions further enhances practical usability by generating natural language treatment recommendations, bridging the gap between technical predictions and actionable agricultural decisions.

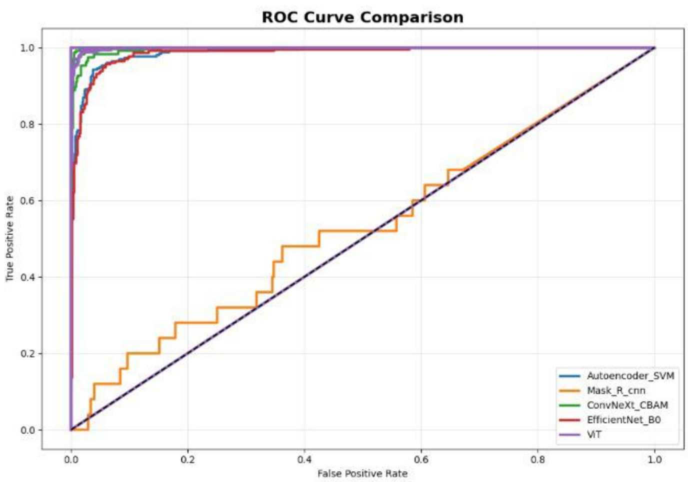

Figure 28: ROC Curve Comparison across All Models

8. Key Takeaways and Lessons Learned

Key Insight from Results Model Comparison

The experimental evaluation demonstrates that all investigated architectures, including EfficientNet-B0, ConvNeXt, Mask R-CNN, and ViT, achieved consistently strong performance, with classification accuracies ranging between 93% and 97%. This indicates that all models were capable of effectively learning discriminative representations from the dataset without any significant underperformance.

Despite this overall high performance, none of the convolution-based or segmentation-based architectures exceeded the performance of the Vision Transformer. Specifically, ViT consistently achieved the highest accuracy of approximately 97%, representing the upper bound among all evaluated models.

Result Interpretation

The observed superiority of ViT can be attributed to its self-attention mechanism, which enables the modeling of long-range dependencies and global contextual relationships within images. In contrast, convolutional architectures such as EfficientNet-B0 and ConvNeXt primarily focus on local feature extraction, which may limit their ability to capture global spatial interactions.

Nevertheless, the performance margin between ViT and CNN-based models remains relatively small, suggesting that convolutional architectures still provide highly competitive feature learning capabilities for this task.

Surprisingly, although Vision Transformer achieved the best overall performance, the improvement over CNN-based architectures was smaller than initially expected. This suggests that modern convolutional models remain highly competitive for medium-scale image classification tasks.

Key Takeaway (Generalizable Insight)

The results highlight that while transformer-based architectures may achieve marginally superior predictive performance, this improvement must be evaluated in conjunction with computational complexity and deployment feasibility. Consequently, model selection in real-world applications should consider not only accuracy but also efficiency, scalability, and resource constraints, particularly in edge or resource-limited environments.

For practitioners, this indicates that selecting the most complex architecture is not always necessary. In scenarios with limited computational resources or real-time deployment requirements, lightweight CNN-based models may provide a more practical balance between accuracy and efficiency.

9.Conclusion & Future Work

9.1 Conclusion

This project focused on grape leaf disease classification using deep learning and machine learning. Several models were implemented, including EfficientNet-B0 efficient and accurate, Autoencoder + SVM a hybrid approach for feature extraction and classification, ConvNeXt-CBAM for focusing on disease-relevant features, and Mask R-CNN for detection and segmentation.

ViT achieved the highest accuracy 98.52% but had high computational complexity. Autoencoder + SVM showed the fastest inference time 0.02s but had lower accuracy.

EfficientNet-B0 provided a good balance between performance and efficiency, making it suitable for resource-limited environments. ConvNeXt-CBAM demonstrated strong accuracy 97.17% and a good balance between performance and computational cost, while Mask R-CNN achieved 95.50% accuracy.

Among the models, EfficientNet-B0 was found to be the most practical for resource-constrained environments, offering a balance of performance and efficiency.

9.2 Recommendations

For plant disease classification, we recommend using deep learning models like ViT or ConvNeXt-CBAM for the best accuracy, provided sufficient computational resources. Autoencoder + SVM should be avoided for complex tasks due to its lower accuracy. For multi-solution approaches, starting with EfficientNet-B0 or ConvNeXt-CBAM is ideal, with time allocated for hyperparameter tuning and preprocessing.

9.3 Future Work and Open Questions

Future work can explore models like VT or DeepLabV3 for improved disease segmentation. Expanding the dataset with more labeled images or synthetic data will improve model generalization. Advanced preprocessing techniques and hyperparameter optimization using AutoML can further enhance model performance.

Enhancing model interpretability and considering real-time deployment on mobile devices or edge platforms will improve the practical application of these systems in agriculture.

10. Project Challenges and Risk Mitigation

The project faced several challenges, particularly with data quality, as the “Healthy Grape” class was underrepresented in the dataset. To address this, we used data augmentation techniques like rotation and flipping to balance the dataset and reduce overfitting. Model complexity was another obstacle, especially with ViT and ConvNeXt-CBAM, which required significant computational resources. We mitigated this by using transfer learning and leveraging cloud resources to reduce training time and computational load. Limited computational resources also posed a challenge, but we used GPU acceleration and cloud platforms to handle the heavy models efficiently.

Implementation difficulties included issues with the segmentation masks not matching the color images. This was addressed by adjusting the preprocessing pipeline to ensure alignment and consistency across the dataset.

Managing four parallel solutions was challenging, but we overcame this by holding regular team meetings and using a project management platform to track progress and ensure alignment. Communication issues were mitigated through Slack for quick communication and Google Docs for real-time collaboration.

The initial timeline was overestimated due to the complexity of training. To address this, we adjusted the schedule and prioritized models that required less training time. Preprocessing took longer than expected, but we automated parts of the process and distributed the workload for hyperparameter tuning across the team. Resource limitations were also a challenge, but we used cloud platforms to ensure smooth progress.

We addressed these challenges by using cloud resources for computational power, ensuring team coordination, and utilizing pre-trained models and transfer learning. We may need additional guidance on hyperparameter optimization and handling large datasets and complex models like ViT and Mask R-CNN.

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